Elif Bilge

bbilge.elif@gmail.com | bilgeelif/projects | linkedin.com/in/elif-bilge | github.com/bilgeelif

About Me

Computer Vision Research Engineer with 4 years of experience developing AI systems for medical imaging. Led research and project initiatives, including grant writing, literature reviews, and technical feasibility analyses, with a track record of delivering production-ready solutions. Translated clinical requirements into optimized solutions through collaboration with clinicians and academic partners. Visit my for website for detailed project portfolio.

Experience

Computer Vision Research Engineer, Disun Software – Remote March 2022 - Oct 2025

Built end-to-end deep learning pipelines for TÜBİTAK R&D projects, developing custom 2D/3D architectures for CT, MRI, and multi-modal imaging. The solutions were integrated into radiology workflows at the collaborating hospitals.

- MLN Comprehensive Analysis: Designed and implemented an algorithm for lymph node segmentation, detection, and station-malignancy classification, providing radiologists with a comprehensive decision-support system that enhances cancer staging accuracy and reduces workload.
- Tumor Budding Segmentation: Developed a neural network pipeline for tumor budding segmentation in H&E images, using multi-resolution ROI analysis to reduce false positives and deployed via a user-friendly interface for pathologist evaluation.
- HE2DAPI Multi-Modal Image Registration: Implemented a keypoint-based image registration pipeline for H&E and DAPI whole-slide images, enabling precise spatial alignment that improved cell type identification, tissue morphology analysis, and diagnostic accuracy, with an interactive GUI for pathologist evaluation.

Research Engineer, UMRAM/Bilkent University – Ankara, TR Feb 2021 – Dec 2022

- MRI Spectrometer Development: Conducted foundational research and authored the project proposal for a national FPGA-based MRI development initiative in collaboration with ASELSAN, aimed at reducing vendor dependency.

Computer Vision Research Engineer, Papilon Defense – Ankara, TR July 2020 – Nov 2020

- License Plate Recognition System: Designed and deployed a real-time license plate detection and OCR pipeline for smart parking systems, combining a YOLO-based detector with custom-trained character recognition model to ensure robust, high-accuracy performance.

Electronic Engineer, Intern, ASELSAN – Ankara, TR Jan 2018 – Apr 2018

- Web Service Architecture Development: Implemented secure user authentication system using Windows Communication Foundation (WCF) framework with MVC design pattern.

Electronic Engineer, Intern, KIWI – Ankara, TR July 2017 – Aug 2017

- Drone Motor Control System (FPGA, VHDL): Developed FPGA-based communication protocol for precision drone motor control systems.

Education

Bilkent University, B.S. in Electrical and Electronics Engineering Sept 2015 – Feb 2020
ODTU, M.S. in Medical Informatics (Terminated by my decision) Terminated by my decision

Technical Skills

Programming Languages: Python, MATLAB, C++, SQL
ML/DL libraries: PyTorch, TensorFlow/Keras, Scikit-Learn, OpenCV, Scipy, NumPy, Pandas, SimpleITK, PyDicom, ITK-SNAP, 3D Slicer, MONAI, HuggingFace
Tools & Platforms: Linux, Docker, Git, WandB, MLflow, CUDA
Languages: English (C1), Turkish (Native)